

A - Maximal Value

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

Problem Statement

You are given an integer sequence of length N : $A = (A_1, A_2, \dots, A_N)$.

Find the number of integers i between 1 and $N - 2$ (inclusive) satisfying $A_i < A_{i+1} > A_{i+2}$.

Constraints

- $3 \leq N \leq 100$
- $1 \leq A_i \leq 100$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
A_1 A_2 ... A_N
```

Output

Output the answer.

Sample Input 1

```
6
3 1 4 1 5 2
```

Sample Output 1

```
2
```

Two values $i = 2, 4$ satisfy the condition. Thus, output 2.

Sample Input 2

```
5
1 1 1 2 1
```

Sample Output 2

```
1
```

Sample Input 3

```
10
7 3 9 8 10 3 1 5 5 4
```

Sample Output 3

```
2
```

B - Corridor Watch

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 200 points

Problem Statement

You are given integers M , D and a string S of length M consisting of G and ..

There are M cells arranged in a row from left to right, numbered 1 through M from the left.

Some of the cells have a guardman standing on them. Specifically, a guardman stands on cell i if $S_i = \text{G}$, and no one stands on cell i if $S_i = \text{.}$.

A cell whose distance from a cell with a guardman is at most D is watched by that guardman. That is, a cell x is watched by a guardman if there exists a cell i such that $S_i = \text{G}$ and $|x - i| \leq D$.

Among the M cells, find the number of cells that are **not** watched.

Constraints

- $0 \leq D < M \leq 100$
- D and M are integers.
- S_i is a string of length M consisting of G and ..

Input

The input is given from Standard Input in the following format:

```
 $M$   $D$   
 $S$ 
```

Output

Output the answer.

Sample Input 1

```
7 1  
.G...GG
```

Sample Output 1

1

Only cell 4 is not watched.

Sample Input 2

6 5
.....

Sample Output 2

6

All cells are not watched.

Sample Input 3

21 2
....G...GG.....G.....

Sample Output 3

6

C - Between P and Q

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

Problem Statement

You are given an integer N and integer sequences $P = (P_1, P_2, \dots, P_N)$ and $Q = (Q_1, Q_2, \dots, Q_N)$, each of which is a permutation of $(1, 2, \dots, N)$.

Find the number of integer sequences that are a permutation of $(1, 2, \dots, N)$ and are lexicographically greater than P and lexicographically less than Q .

► What is lexicographic order for integer sequences?

Constraints

- $1 \leq N \leq 10$
- P and Q are integer sequences that are permutations of $(1, 2, \dots, N)$.
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
P1 P2 ... PN
Q1 Q2 ... QN
```

Output

Output the answer.

Sample Input 1

```
3
1 3 2
3 1 2
```

Sample Output 1

```
2
```

Two sequences $(2, 1, 3)$, $(2, 3, 1)$ satisfy the condition. Thus, output 2.

Sample Input 2

```
5
5 4 2 1 3
5 1 2 3 4
```

Sample Output 2

```
0
```

There is no permutation of $(1, 2, 3, 4, 5)$ satisfying the condition.

Sample Input 3

```
7
3 6 5 2 7 1 4
4 1 5 7 2 3 6
```

Sample Output 3

```
223
```

D - Pre-Palindrome

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 400 points

Problem Statement

A string consisting of lowercase English letters is called a **good string** if it satisfies the following condition.

- It can be turned into a palindrome by rewriting at most one character.

For example, a, iwai, and abcdcza are good strings, but abcd and atcoder are not good strings. Note that, in particular, a palindrome is also a good string.

You are given a string S consisting of lowercase English letters.

Find the number of non-empty substrings (contiguous subsequences) of S that are good strings.

Two substrings taken from different positions of S are counted separately even if they are equal as strings.

► What is a substring?

Constraints

- S is a string of length between 1 and 10^4 , inclusive, consisting of lowercase English letters.

Input

The input is given from Standard Input in the following format:

S

Output

Output the answer.

Sample Input 1

ababa

Sample Output 1

13

The substring `ba` from the 2nd through 3rd characters of S and the substring `ababa` from the 1st through 5th characters are some good strings.

Among the 15 substrings, the good strings are the 13 substrings other than `abab` from the 1st through 4th characters and `baba` from the 2nd through 5th characters.

Sample Input 2

atcoder

Sample Output 2

18

Sample Input 3

abccbacbacb

Sample Output 3

40

E - Sum of Average

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 450 points

Problem Statement

You are given a positive integer N and a length- N integer sequence $A = (A_1, A_2, \dots, A_N)$.

Define $f(l, r)$ as the (arithmetic) mean of $A_l, A_{l+1}, \dots, A_r$.

Find $\sum_{1 \leq l \leq r \leq N} f(l, r)$, modulo 998244353.

► Definition of a rational number modulo 998244353

Constraints

- $1 \leq N \leq 5 \times 10^5$
- $0 \leq A_i < 998244353$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
 $N$   
 $A_1$   $A_2$   $\dots$   $A_N$ 
```

Output

Output the answer.

Sample Input 1

```
2  
2 3
```

Sample Output 1

499122184

We have $f(1, 1) = 2, f(1, 2) = \frac{2 + 3}{2} = \frac{5}{2}, f(2, 2) = 3$. Thus, $\sum_{1 \leq l \leq r \leq N} f(l, r) = 2 + \frac{5}{2} + 3 = \frac{15}{2}$.
 $\frac{15}{2}$ in modulo-998244353 expression is 499122184, so output 499122184.

Sample Input 2

6
1 2 3 4 5 6

Sample Output 2

499122250

Sample Input 3

9
3 1 4 1 5 9 2 6 5

Sample Output 3

855638200

F - Chmax

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 500 points

Problem Statement

You are given a positive integer N and a permutation $P = (P_1, P_2, \dots, P_N)$ of $(1, 2, \dots, N)$.

There are variables x, y, c . Initially, $x = y = c = 0$.

For $k = 1, 2, \dots, N$ in this order, you perform one of the following operations:

- Operation 1: Increase c by 1 if $x < P_k$. Then, replace x with $\max(x, P_k)$.
- Operation 2: Increase c by 1 if $y < P_k$. Then, replace y with $\max(y, P_k)$.

Find the maximum possible final value of c .

Constraints

- $1 \leq N \leq 5 \times 10^5$
- P is a permutation of $(1, 2, \dots, N)$.
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
P_1 P_2 ... P_N
```

Output

Output the answer.

Sample Input 1

```
5
4 3 1 2 5
```

Sample Output 1

4

By performing the operations as follows, you can achieve $c = 4$:

- For $k = 1$: Perform operation 1. Then, $(x, y, c) = (4, 0, 1)$.
- For $k = 2$: Perform operation 1. Then, $(x, y, c) = (4, 0, 1)$.
- For $k = 3$: Perform operation 2. Then, $(x, y, c) = (4, 1, 2)$.
- For $k = 4$: Perform operation 2. Then, $(x, y, c) = (4, 2, 3)$.
- For $k = 5$: Perform operation 2. Then, $(x, y, c) = (4, 5, 4)$.

c cannot be made greater than 4 no matter how you perform the operations, so output 4.

Sample Input 2

6
6 5 4 3 2 1

Sample Output 2

2

Sample Input 3

9
3 6 5 2 7 8 9 1 4

Sample Output 3

7

G - Restricted Permutation

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 550 points

Problem Statement

You are given an integer N and a string S of length N consisting of o and x.

Find the number, modulo 998244353, of permutations $P = (P_1, P_2, \dots, P_N)$ of $(1, 2, \dots, N)$ satisfying the following condition.

- For $k = 1, 2, \dots, N$, the following two are equivalent.
 - $S_k = \text{o}$
 - P contains a permutation of $(1, 2, \dots, k)$ as a contiguous subsequence.

Constraints

- $1 \leq N \leq 2000$
- S_i is a string of length N consisting of o and x.

Input

The input is given from Standard Input in the following format:

```
 $N$   
 $S$ 
```

Output

Output the answer.

Sample Input 1

```
3  
oxo
```

Sample Output 1

2

$P = (1, 3, 2), (2, 3, 1)$ satisfy the condition.

Sample Input 2

7
XXXXXXX

Sample Output 2

0

There is no P satisfying the condition.

Sample Input 3

15
OXXXOXXXXXOXXO

Sample Output 3

1627648

Hints: AtCoder Beginner Contest 468

Taken from the official editorial, not written by us. Read one hint at a time.

A. Maximal Value *(editorial, first line)*

Count the number of indices i satisfying the condition.

B. Corridor Watch *(editorial, first line)*

A cell is not watched if and only if there is no guardman on a cell distant by at most D .

C. Between P and Q *(editorial, first line)*

There are $N!$ permutations of $(1, 2, \dots, N)$.

D. Pre-Palindrome *(editorial, first line)*

First, we will count the number of odd-length good strings.

E. Sum of Average *(editorial, first line)*

For an integer n , let $H_n = \sum_{k=1}^n \frac{1}{k}$.

F. Chmax *(editorial, first line)*

First of all, consider the DP (Dynamic Programming) where $d[i][p][q]$ is defined as the maximum value of c when the operations up to $k = i$ have finished, where $x = p$ and $y = q$.

G. Restricted Permutation *(editorial, first line)*

First of all, the permutation P always contain permutations of (1) and $(1, 2, \dots, N)$ as a subarray, so the answer is 0 if $S_1 = x$ or $S_N = x$.

Source: <https://atcoder.jp/contests/abc468/editorial>